Supplementary Material: Conchordal: Emergent Harmony via Direct Cognitive Coupling in a Psychoacoustic Landscape

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S1. Algorithm Pseudocode

S1.1 Consonance Search

Input: agents $\{a_1, \ldots, a_N\}$ with pitches x_i in $\log_2$-freq space.
Parameters: step size $\lambda = 25\,\text{ct}$, crowding weight $\lambda_C = 0.15$, $T_0 = 0.05$, $\tau = 4.8$ sweeps, 8 sweeps total, burn-in $= 2$, phase switch at sweep 4.

`for` sweep $t = 0$ `to` 7:
  `if` $t = 4$: reset phase-local annealing clock
  $T_t \leftarrow T_0 \cdot \exp(-t_{\text{phase}}/\tau)$
  `for` each agent a_i:
    $\delta \sim \text{Uniform}(-\lambda, +\lambda); \quad x' \leftarrow x_i + \delta$
    $P(x) = \lambda_C \sum_{j \neq i} K_{\text{crowd}}(\Delta_{\text{erb}}(x, x_j))$
    $S_{\text{old}} \leftarrow s(t) \cdot C^{\text{LOO}}_{\text{field}}(x_i) - P(x_i)$
    $S_{\text{new}} \leftarrow s(t) \cdot C^{\text{LOO}}_{\text{field}}(x') - P(x')$
    `if` $U(0,1) < \min\big(1, \exp\big((S_{\text{new}} - S_{\text{old}})/T_t\big)\big)$ `then` $x_i \leftarrow x'$

where $s(t) = -1$ for $t < 4$ (exploration / anti-clustering phase), $+1$ for $t \geq 4$ (consonance-seeking phase); statistics are computed from sweeps $2 \ldots 7$ only.

S1.2 Selection Pressure

Parameters: $c_b = 0.5$/s, $r_E = 0.4$/s, $\Delta t = 512/48000\,\text{s}$.

`for` each time step:
  `for` each agent a_i:
    $C_{\text{score}} \leftarrow$ raw reference-landscape consonance at x_i
    $E_i \leftarrow E_i - c_b \cdot \Delta t$ *(basal drain)*
    $E_i \leftarrow E_i + r_E \cdot \max(0, C_{\text{score}}) \cdot \Delta t$ *(continuous recharge)*
    `if` $E_i \leq 0$: agent dies; respawn at random pitch within ± 1 oct

S1.3 Temporal Scaffold

Parameters: $k_{\text{time}} = 3.0$, kick at 2 Hz, $\omega_0 = 1.8 \cdot 2\pi$ rad/s, $\pm 2\%$ jitter.

`for` each time step dt:
  Advance canonical beat phase $\theta(t)$ at 2 Hz
  `if` condition = scrambled `and` cycle boundary crossed: sample new offset $\psi \sim \text{Uniform}(0, 2\pi)$
  `for` each agent a_i:
    Set drive phase $\theta_i^* \leftarrow \theta(t)$ (shared), $\theta(t) + \psi$ (scrambled), or none (off)
    $k_{\text{eff},i} \leftarrow k_{\text{time}}$ (shared/scrambled) or 0 (off)
    $\dot{\phi}_i \leftarrow \omega_i + (k_{\text{eff},i}/N) \sum_j \sin(\phi_j - \phi_i) + F(\theta_i^*)$
    $\phi_i \leftarrow \phi_i + \dot{\phi}_i \cdot dt$
  Record PLV and onset phase relative to the canonical beat $\theta(t)$

S1.4 Minimal Hereditary Adaptation

Key parameters: $N = 16$, range ± 2 oct, candidate count $K = 16$, slot-local settling $= 2$ ticks, family-local azimuth search radius 0.5 st, survival thresholds $(0.30, 0.80)$.

`for` each time step:
  Advance metabolism using the polyphonic leave-one-out score (C_{score} with crowding/capacity penalties)
  `if` selection = on:
    $s_i \leftarrow \text{clamp}\big((C_{\text{score},i} - 0.30)/(0.80 - 0.30),\ 0,\ 1\big)$
    $E_i \leftarrow E_i - c_b \Delta t + r_E\, s_i\, \Delta t$
    Parent-choice weights depend on current score and energy
  `if` selection = off: use neutral turnover with a small background-death floor
  `if` agent a_i dies:
    `if` respawn = heredity:
      Choose parent a_p from survivors
      Form a parent-biased set of harmonic-family peaks under the shared scene filter
    `else` (matched random control):
      Sample K log-random candidates across tessitura and score them with the same scene filter
    Choose one target family / slot
    Search locally within the chosen family to set the final azimuth
    Respawn with initial energy E_0

Matched random control: adult pitch remains locked, the same metabolic policy is used, and the same short slot-local settling phase is applied; only the parent-biased family choice is removed.

S2. Normalization of H_{01} and R_{01}

Both harmonicity and roughness are normalized to $[0, 1]$ before entering the consonance core.

Harmonicity (H_{01})

The raw harmonicity potential from sibling projection (Eqs. 1–2 in the main text) is normalized by dividing by the self-harmonicity of a single harmonic tone at the reference position (which equals the kernel's maximum response):

$$H_{01}(x) = \mathrm{clamp}\left(\frac{H_{\mathrm{raw}}(x)}{H_{\mathrm{ref,max}}},\ 0,\ 1\right), \tag{S1}$$

where $H_{\mathrm{ref,max}} = 1.0$ (the kernel is pre-normalized so that a single harmonic stack produces unit peak).

Roughness (R_{01})

Raw roughness is first expressed as a ratio $\rho = R_{\mathrm{raw}}/R_{\mathrm{ref}}$, where R_{ref} is the roughness of a single-tone reference density (computed once at initialization). This ratio is then mapped through a piecewise saturation function with physiological constant k ($k = 1.0$ in all experiments):

$$R_{01}(\rho) = \begin{cases} 0 & \rho \le 0, \\ \dfrac{\rho}{1+k} & 0 < \rho < 1, \\ 1 - \dfrac{k}{\rho + k} & \rho \ge 1, \end{cases} \tag{S2}$$

clamped to $[0, 1]$. For $\rho < 1$ (below-reference roughness) the mapping is linear; for $\rho \ge 1$ it is a hyperbolic saturation that approaches 1 asymptotically as $\rho \to \infty$. At $\rho = 1$ the function is continuous: $R_{01}(1) = 1/(1 + k)$. With $k = 1$, the reference roughness maps to $R_{01} = 0.5$.

S3. Complete Parameter Table

Table S1 lists all parameters used across experiments.

S3.1 Supplementary Mechanism Check: Configurable Spectral–Temporal Bridge

The relation between Conchordal's spectral and temporal axes is not treated as a uniquely determined neurocognitive law. Existing neuroscience motivates several plausible couplings between harmonic structure, salience, attention, and rhythmic prediction, but does not reduce them to a single fixed mapping. Conchordal therefore treats the bridge from spectral state to temporal coupling as a design degree of freedom available to the composer. The current implementation includes one such mapping: consonance modulates metabolic vitality, and vitality gates the effective coupling strength of the phase oscillator. In the supplementary assay shown here, the bridge uses $R = 1.0$/s, $\gamma = 0.5$/s, $E_{\mathrm{cap}} = 1.0$, $\lambda_v = 1.0$, and $v_{\mathrm{floor}} = 0.0$. Figure S1 reports this assay as an implementation-level mechanism check, not as evidence that consonance uniquely determines rhythmic coupling in biological cognition.

S4. Minimal Hereditary Adaptation: Current Supplementary Baselines

The current paper-facing hereditary assay is no longer the legacy parameter sweep reported in earlier drafts. The main text uses one fixed design and compares heredity against a *matched log-random* baseline that shares the same scene filter and short slot-local settling phase. Table S2 reports the current selection-on comparisons for the paper assay and two stronger supplementary random baselines. These additional baselines are useful diagnostically, but they are not equivalent controls for musical polyphony: the unrestricted hard-random restart can maximize raw C_{score} by collapsing into very few occupied bands.

The main hereditary claim is therefore intentionally conservative. It is made against the matched log-random baseline rather than against the unrestricted hard-random restart, because the latter is a much stronger optimizer and tends to over-concentrate into a few occupied bands. Under the matched baseline, the current assay supports the paper's mechanistic claim: lineage-biased family choice plus local azimuth search plus metabolic selection yields more musical, less diffuse polyphony than matched random respawn. The full 20-seed rerun also preserves the trajectory advantage reported in the main text ($\mathrm{AUC}_C = 0.510$ vs. 0.350) and the threshold-crossing asymmetry (15/20 heredity+selection seeds reach $\mathrm{JI}_{\mathrm{score}} \ge 0.50$, vs. 0/20 matched-random+selection seeds).

S5. Bilinear Consonance Core: Derivation

The consonance core (Eq. 5 in the main text) is the most general bilinear form for two inputs $H_{01}, R_{01} \in [0, 1]$:

$$C = a\,H_{01} + b\,R_{01} + c\,H_{01}\,R_{01} + d. \tag{S3}$$

$C_{\textbf{field}}$ $(a, b, c, d) = (1, -1.35, 1, 0)$:

$$C_{\mathrm{field}} = H_{01} - 1.35\,R_{01} + H_{01}\,R_{01} \tag{S4}$$

$$= H_{01}(1 + R_{01}) - 1.35\,R_{01}. \tag{S5}$$

When H_{01} is high (strong harmonicity), the $H_{01}\,R_{01}$ term partially compensates for roughness, reflecting the perceptual observation that harmonically rich intervals tolerate moderate beating. When $H_{01} = 0$, roughness is penalized at full strength.

$C_{\textbf{density}}$ $(a, b, c, d) = (1, 0, -1, 0)$:

$$C_{\mathrm{density}} = H_{01}(1 - R_{01}). \tag{S6}$$

This product is nonnegative and peaks where harmonicity is high and roughness is low. In Conchordal's interactive mode it serves as a spawn-position weight; it is not used in the experiments reported in this paper but is shown in Fig. 1D for comparison.

$C_{\textbf{level01}}$: C_{field} passed through a logistic sigmoid $\sigma(\beta(x - \theta))$ with $\beta = 2$, $\theta = 0$. In the current paper it is used for reporting and supplementary bridge variants; it is not the direct

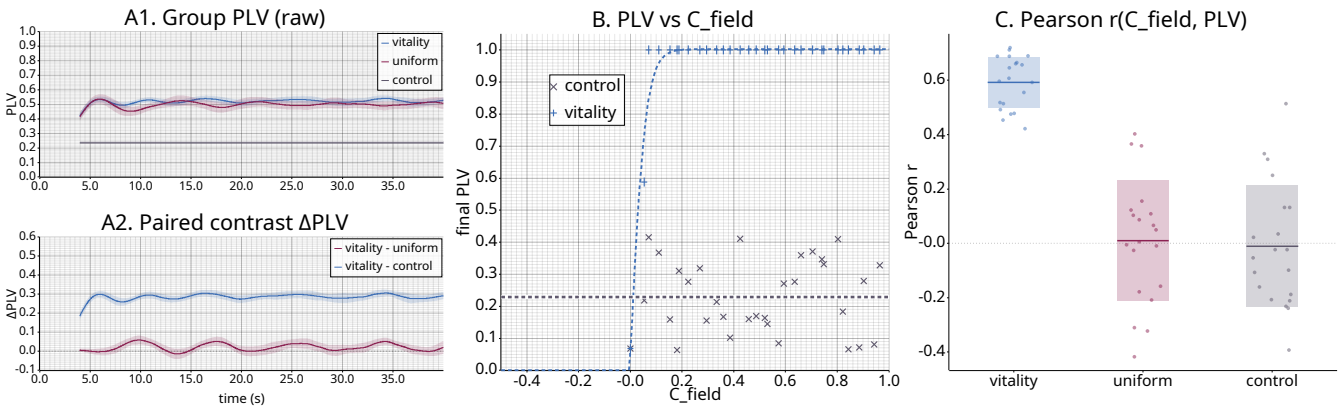

Figure S1: Supplementary mechanism check for one configurable spectral–temporal bridge (20 seeds $\times$ 3 conditions). (A1) Group mean PLV over time (95% CI). (A2) Paired contrast ΔPLV (95% CI). (B) Per-agent final PLV vs. C_{field} (representative seed) with sigmoid fit (c_{50} fixed at 0). (C) Seed-level Pearson $r(C_{\text{field}}, \text{PLV})$. In the current implementation, vitality-gated coupling yields a graded bridge from spectral state to entrainment strength; alternative bridge mappings remain possible within the broader Conchordal design.

recharge law in either the selection-pressure or hereditary assays.

S6. Statistical Status of the Current Hereditary Assay

Earlier versions of this project reported a 20-seed factorial heredity analysis with an exact sign-flip permutation test. That legacy test no longer matches the present paper-facing hereditary assay and has therefore been removed from the main argument. The current Fig. 4 instead uses a full 20-seed rerun of the revised design, and the paper reports seed-level means, 95% confidence intervals, trajectory area (AUC_C), and threshold-crossing counts rather than reusing the obsolete legacy permutation result.

The current hereditary result should therefore be read as a full seed-level comparison for the revised mechanism, but not yet as the final word on inferential robustness across all conceivable terrain and topology variants. A renewed non-parametric test could be added in future work if the present assay is frozen as the final reference design.

S7. Consonance-Only Control

The consonance-search assay uses a two-phase curriculum: an exploration / anti-clustering phase (sweeps 0–3) followed by a consonance-seeking phase (sweeps 4–7). To assess whether the resulting polyphony depends on this schedule, we ran a *consonance-only* control with identical parameters but $s(t) = +1$ for all 8 sweeps (*local-search* condition, 20 seeds).

Both conditions converge to high-consonance states, but they are not interchangeable. The consonance-only control ends slightly higher on raw C_{score} (0.745 vs. 0.676; $p < 0.001$), while the curriculum produces a more spacious endpoint: more unique bins (11.9 vs. 8.4; $p < 0.001$) and wider nearest-neighbour spacing (0.576 vs. 0.250 st; $p < 0.001$). Under the same primary endpoint metric used in the main text, interval entropy is nearly unchanged (2.013 vs. 2.045 nats; $p = 0.696$). In the current 8-sweep assay, the pre-switch phase therefore acts less as a route to higher final consonance than as a route to broader basin coverage and better polyphonic spacing.

S7.1 Derived Metrics: LOO Score and Scene Consonance

Leave-One-Out (LOO) Score. For each agent i, the field spectrum is recomputed with agent i's harmonic contribution removed, yielding a leave-one-out field F_{-i}. The agent's LOO consonance is the consonance field evaluated at its own position x_i against the remaining agents:

$$C_{\text{score},i} = C_{\text{field}}(F_{-i},\, x_i). \tag{S7}$$

The population mean defines the primary outcome metric:

$$C_{\text{score}} = \frac{1}{N}\sum_{i=1}^{N} C_{\text{score},i}. \tag{S8}$$

Scene Consonance $G(F)$. Scene harmonicity H_{soc} and roughness R_{soc} quantify the *emergent* harmonic and roughness structure beyond what individual agents would produce in isolation. Let $H(F)$ and $R(F)$ denote harmonicity and roughness evaluated on the full scene spectrum F, and let $H(\{i\})$ and $R(\{i\})$ denote the same quantities for agent i

presented alone. Then:

$$H_{\text{soc}} = H(F) - \frac{1}{N}\sum_{i=1}^{N} H(\{i\}), \tag{S9}$$

$$R_{\text{soc}} = R(F) - \frac{1}{N}\sum_{i=1}^{N} R(\{i\}). \tag{S10}$$

Scene consonance applies the same bilinear core (Eq. 5 in the main text):

$$G(F) = a\,H_{\text{soc}} + b\,R_{\text{soc}} + c\,H_{\text{soc}}\,R_{\text{soc}}. \tag{S11}$$

By subtracting the mean singleton contribution, $G(F)$ isolates the consonance arising from inter-agent spectral interactions rather than individual tonal quality.

S8. Terrain Control and Robustness Analysis

Three supplementary analyses test whether the Consonance Search result depends on the specific structure of the psychoacoustic landscape rather than on incidental parameter choices.

S8.1 Shuffled-Landscape Control

We destroy spatial coherence by applying a fixed random permutation of the C_{score} bin indices, generated once per run and held fixed throughout all sweeps. Agents therefore move on a stationary landscape with the same *marginal* score distribution but no local gradient structure. Table S4 compares the shuffled condition with the standard *local-search* condition ($n = 20$ seeds each).

C_{score} (ref) denotes the consonance score of each final configuration evaluated on the *original* (unshuffled) landscape, enabling a fair cross-condition comparison. Without coherent spatial gradients, crowding alone over-disperses the population: unique bins rise from 11.9 to 17.0 and nearest-neighbour spacing widens from 0.576 to 0.928 st (both $p < 0.001$), but interval structure degrades sharply: entropy rises from 2.013 to 4.416 nats and C_{score} falls from 0.676 to 0.623 ($p \leq 0.001$). The coherent psychoacoustic landscape is therefore necessary for agents to self-organise into musically structured pitch configurations.

S8.2 Harmonicity–Roughness Misalignment Control

The consonance kernel combines harmonicity (H_{01}) and roughness (R_{01}). We test whether their spatial alignment matters by circularly shifting R_{01} by Δ cents before scoring, which preserves each component's marginal statistics but breaks the psychoacoustically motivated registration between them. Four shift values span the range $\Delta \in \{137, 271, 389, 523\}$ cents. Table S5 reports the results ($n = 20$ seeds per shift, 80 runs total for the misaligned aggregate).

All four misaligned conditions produce significantly fewer unique pitch bins than the local-search baseline (all $p \leq 0.0025$, Welch t), dropping from 11.9 to about 9 bins. Nearest-neighbour distances also decrease significantly (all $p \leq 0.0001$), and entropy rises from the baseline 2.013 to 2.18–2.26 nats. Misalignment leaves the final reference-evaluated C_{score} broadly intact (0.681–0.701 vs. 0.676 baseline) while degrading spacing and interval structure. The alignment between harmonicity and roughness therefore matters less for raw endpoint consonance than for the formation of a differentiated, well-spaced polyphonic distribution.

S8.3 Consonance Kernel Coefficient Sensitivity

The consonance kernel $C = a\,H_{01} + b\,R_{01} + c\,H_{01}\,R_{01} + d$ has four coefficients; the default is $(a, b, c, d) = (1.0, -1.35, 1.0, 0.0)$. We sweep $b \in \{-2.0, -1.35, -0.7, 0.0\}$ and $c \in \{0.0, 0.5, 1.0, 1.5\}$ with $a = 1$, $d = 0$ fixed (16 conditions, 20 seeds each). Table S6 reports the results; because the sweep varies the kernel while holding everything else constant, it isolates the sensitivity of the consonance landscape to its parameterisation.

Unique-bin counts (7.8–9.6) and entropy (1.98–2.30 nats) remain within a fairly narrow band across the entire (b, c) grid, showing that the emergent pitch structure is not fragile to kernel parameterisation. C_{score} increases when the roughness penalty $|b|$ decreases or the interaction weight c increases (0.687–0.786), which is expected from the kernel definition. However, none of the coefficient settings reproduce the full curriculum baseline's 11.9 unique bins, confirming that the curriculum and landscape coherence—not coefficient fine-tuning alone—are responsible for the broader, better-spaced polyphony of the main consonance-search condition.

S8.4 Terrain Control (Minimal Hereditary Adaptation)

The updated hereditary assay has *not* yet been rerun on shuffled terrain. The corresponding legacy table was removed because it described an earlier design that no longer matches the current paper-facing hereditary mechanism. The present hereditary claim should therefore be read as mechanism-level evidence within one well-characterized psychoacoustic terrain, exactly as stated in the main text.

S8.5 Topology Generalization (Minimal Hereditary Adaptation)

Likewise, the updated hereditary assay has not yet been rerun across the partial-count topology battery used for the consonance-search assay. No topology-generalization table is therefore reported here. When this battery is rerun, it will use the current paper-facing hereditary mechanism: family-biased respawn, local azimuth search, metabolic selection, and the matched log-random baseline.

S8.6 Phase Coherence (Minimal Hereditary Adaptation)

Phase-coherence readouts from earlier hereditary builds are not carried forward into the present paper-facing assay. The current hereditary claim is intentionally spectral and lineage-based: it concerns where offspring are placed in pitch space and how that placement accumulates across generations. Any future reintroduction of rhythmic-order statistics for the hereditary assay should be treated as a separate supplementary extension rather than as part of the current main claim.

Table S1: Complete parameter listing for all experiments.

Parameter	Symbol	Consonance Search	Selection	Temporal Scaffold	Hereditary Adaptation
Anchor frequency	f_0	220 Hz	220 Hz	220 Hz	220 Hz
Population size	N	24	32	32	16
Seeds		20	20	20	20
Pitch range		± 2 oct	± 1 oct	± 1 oct	± 2 oct
Harmonicity / Roughness					
Harmonic order	N_H	16	16	16	16
Power-law exponent	ρ	0.4	0.4	0.4	0.4
Roughness saturation	k	1.0	1.0	1.0	1.0
Consonance core					
C_{field} coefficients	(a, b, c, d)	$(1, -1.35, 1, 0)$	$(1, -1.35, 1, 0)$	$(1, -1.35, 1, 0)$	$(1, -1.35, 1, 0)$
C_{density} coefficients[‡]	(a, b, c, d)	$(1, 0, -1, 0)$	—	—	—
Auxiliary bounded map[§]		—	$(2, 0)$	—	—
Pitch adaptation					
Step size	λ	0.25 st	—	—	—
Crowding weight	λ_C	0.15	—	—	0.005
Capacity penalty		—	—	—	radius 35 ct; 3 free voices; weight 0.07
Initial temperature	T_0	0.05	—	—	—
Annealing time constant	τ	4.8 sweeps	—	—	—
Duration		8 sweeps	—	—	—
Phase switch		sweep 4	—	—	—
Burn-in discarded		2 sweeps	—	—	—
Metabolism					
Basal cost	c_b	—	0.5/s	—	0.12/s
Recharge coefficient	r_E	—	0.4/s	—	0.20/s (selection-on)
Recharge signal[¶]		—	$\max(0, C_{\text{score}})$	—	$\text{ramp}_{[0.30,0.80]}(C_{\text{score}})$
Energy cap	E_{cap}	—	1.0	—	1.0
Time step	Δt	—	512/48k s	0.02 s	512/48k s
Background death		—	—	—	0.03/s (selection-off only)
Termination		—	200 deaths	40 s	12k-step cap
Minimum deaths		—	—	—	2.5k
Temporal coupling					
Base coupling	k_{time}	—	—	3.0	—
External kick	f_{kick}	—	—	2 Hz	—
Intrinsic frequency	ω_0	—	—	$1.8 \cdot 2\pi$	—
Frequency jitter		—	—	$\pm 2\%$	—
Heredity					
Respawn candidates		—	—	—	matched $K = 16$ in both arms
Parent proposal		—	—	—	contextual-C family peaks
Parent selection		—	—	—	score/energy-weighted (sel. on), uniform (sel. off)
Juvenile settling		—	—	—	2 ticks, 0.20 st radius
Azimuth policy		—	—	—	local search ± 0.5 st at 0.05 st grid
JI cross-validation					
Proximity kernel	α	—	—	—	4.0×10^{-4} ct^{-2}
Ratio limit	p, q	—	—	—	≤ 8
Ratio weight		—	—	—	$1/(pq)$

[‡] Visualization only (Fig. 1D); not used in experiment algorithms.

[§] Bounded auxiliary signal used for reporting / bridge variants; not the hereditary recharge law.

[¶] $\text{ramp}_{[l,h]}(x) = \text{clamp}((x - l)/(h - l), 0, 1)$.

Table S2: Current hereditary-adaptation baselines (selection-on arms, 20 matched seeds). The main paper comparison is heredity+selection vs. matched log-random+selection.

Condition	Baseline type	C_{score}	Entropy	JI_{score}	Unique bins
Heredity+selection	main heredity assay	0.536	3.281	0.323	12.30
Random+selection	matched log-random	0.358	4.220	0.197	14.35
Random+selection	scene-peak matched random	0.365	4.030	0.322	13.40
Random+selection	unrestricted hard random-log	0.845	1.157	0.589	5.75

Table S3: Curriculum vs. consonance-only control (*local-search* condition, $n = 20$ seeds).

Metric	**Curriculum**	**Cons.-only**	t
C_{score} (post)	0.676 ± 0.041	0.745 ± 0.032	5.98^{***}
Unique pitch bins	11.9 ± 1.4	8.4 ± 1.0	-9.19^{***}
NN distance (st)	0.576 ± 0.245	0.250 ± 0.141	-5.15^{***}
Interval entropy	2.013 ± 0.257	2.045 ± 0.261	0.39

Table S4: *Local-search* vs. shuffled landscape.

Metric	**Local-search**	**Shuffled**	t
Unique pitch bins	11.9 ± 1.4	17.0 ± 1.3	11.83^{***}
NN distance (st)	0.576 ± 0.245	0.928 ± 0.161	5.38^{***}
Interval entropy	2.013 ± 0.257	4.416 ± 0.168	35.00^{***}
C_{score} (ref)	0.676 ± 0.041	0.623 ± 0.046	-3.80^{***}

Table S5: *Local-search* vs. H/R-misaligned conditions.

Condition	Unique bins	NN dist. (st)	Entropy
Local-search	11.9 ± 1.4	0.576 ± 0.245	2.013 ± 0.257
$\Delta = 137\,¢$	9.7 ± 1.1	0.292 ± 0.132	2.207 ± 0.244
$\Delta = 271\,¢$	9.1 ± 0.9	0.272 ± 0.136	2.229 ± 0.175
$\Delta = 389\,¢$	9.3 ± 1.1	0.276 ± 0.141	2.264 ± 0.229
$\Delta = 523\,¢$	9.1 ± 0.9	0.241 ± 0.143	2.180 ± 0.210
Misaligned (avg)	9.3 ± 1.0	0.270 ± 0.139	2.220 ± 0.218

Table S6: Coefficient sweep: consonance-search metrics across 16 (b, c) settings.

b	c	Unique bins	Entropy	C_{score}
-2.00	0.00	8.1 ± 0.9	2.069 ± 0.202	0.687 ± 0.047
-2.00	0.50	7.8 ± 0.8	1.981 ± 0.204	0.712 ± 0.038
-2.00	1.00	8.1 ± 1.0	2.033 ± 0.200	0.705 ± 0.043
-2.00	1.50	8.2 ± 0.9	2.048 ± 0.223	0.717 ± 0.035
-1.35	0.00	8.6 ± 0.8	2.098 ± 0.185	0.701 ± 0.041
-1.35	0.50	8.6 ± 1.2	2.057 ± 0.250	0.721 ± 0.042
-1.35	1.00	8.6 ± 1.0	2.125 ± 0.243	0.722 ± 0.032
-1.35	1.50	8.4 ± 1.0	2.054 ± 0.230	0.743 ± 0.033
-0.70	0.00	9.4 ± 1.2	2.239 ± 0.193	0.705 ± 0.031
-0.70	0.50	9.2 ± 1.0	2.222 ± 0.183	0.718 ± 0.028
-0.70	1.00	9.1 ± 1.1	2.193 ± 0.219	0.741 ± 0.032
-0.70	1.50	9.1 ± 1.2	2.142 ± 0.245	0.763 ± 0.047
0.00	0.00	9.6 ± 1.0	2.297 ± 0.202	0.733 ± 0.038
0.00	0.50	9.6 ± 1.0	2.264 ± 0.178	0.746 ± 0.029
0.00	1.00	9.4 ± 1.0	2.168 ± 0.235	0.774 ± 0.026
0.00	1.50	9.4 ± 0.9	2.234 ± 0.220	0.786 ± 0.028